

Chapter 20

The Problem of Moral Hazard

Learning Objectives

- 20.1 Define moral hazard and explain how it is different from adverse selection.
- 20.2 Identify how to anticipate moral hazard and consummate the implied wealth-creating transaction.
- 20.3 Identify solutions to the problem of moral hazard centering on efforts to eliminate information asymmetry.
- 20.4 Describe shirking as a form of moral hazard.
- 20.5 Explain moral hazard in the context of lending.

Chapter 20 – Main Points

- Moral hazard refers to the reduced incentive to exercise care once you purchase insurance.
- Moral hazard occurs in a variety of circumstances: Anticipate it, and (if you can) figure out how to consummate the implied wealth-creating transaction (i.e., ensuring that consumers continue to take care when the benefits of doing so exceed the costs).
- Moral hazard can look very similar to adverse selection—both arise from information asymmetry. Adverse selection arises from hidden information about the type of individual you're dealing with; moral hazard arises from hidden actions.
- Solutions to the problem of moral hazard center on efforts to eliminate the information asymmetry (e.g., by monitoring or by changing the incentives of individuals).
- Shirking is a form of moral hazard.
- Borrowers prefer riskier investments because they get more of the upside while the lender bears more of the downside. Borrowers who have nothing to lose exacerbate this moral hazard problem.

Warm-Up

- Have you ever been in a group project where someone slacked off but still got the same grade as everyone else? What happened next time you worked with them?

Moral Hazard

- **Moral Hazard** is the reduced incentive to exercise care once you purchase insurance.
- It lowers the gain for one party and reduces risks for the other
 - Insurance companies lose money to moral hazard
 - Customers reduce risk because of insurance
- The key is to recognize moral hazard
 - Is one party acting differently than they otherwise would?
 - If insurance reduces care to exercise caution, you have found a moral hazard problem.
- Moral hazard isn't restricted to insurance
 - Worker contracts
 - Worker hours vs. other compensation (% or by the piece)
 - Relationships



Moral Hazard v. Adverse Selection

- **Differentiating Moral Hazard from Adverse Selection**

- Adverse Selection: Information hidden pre-policy (e.g., risk type)
 - Bad drivers are more likely to purchase cars with safety features.
- Moral Hazard: Actions hidden post-policy (e.g., care level)
 - Once individuals gain some form of insurance, they are more likely to take risks.

- **Solutions to Combat Moral Hazard**

- Information gathering by insurance (or other) companies
 - Distinguish between high and low-risk consumers
- Monitoring and rewarding careful behavior

Identifying Moral Hazard vs. Adverse Selection

- Moral hazard arises due to hidden actions
 - i.e., the party partaking in moral hazard hides their true intentions to the other party.
 - e.g., a football player signs a 5-year guaranteed extension after a great season but then doesn't play to expectations.
 - e.g., a driver buys full coverage auto insurance and drives recklessly
 - e.g., a nonsmoker buys the best health insurance policy and suddenly starts smoking.
- Adverse selection arises due to hidden information
 - e.g., the same football player is going through marital problems that affects performance
 - e.g., the same nonsmoker has a history of chronic illness

Activity – Identifying Moral Hazard

Which of the following is an example of moral hazard?

- Reckless drivers are the ones most likely to buy automobile insurance.
- Retail stores located in high-crime areas tend to buy theft insurance more often than stores located in low-crime areas.
- Drivers who have many accidents prefer to buy cars with air bags.
- Employees recently covered by the company health plan start going to the doctor every time they get a cold.

Activity – Identifying Moral Hazard

Which of the following is an example of moral hazard?

- High-quality products being driven out of a market by low quality products.
- A local charity raising insufficient funds because no one contributes, expecting that their neighbors will.
- A bakery defaults on its loan because of a new consumer fear of carbohydrates.
- A corporation uses a business loan secured for one investment on another, higher-risk investment.

Activity – Identifying Moral Hazard

Which of the following is not an example of moral hazard?

- People are more likely to lock their own car than a rental car.
- Skateboarders attempt more difficult maneuvers when wearing a helmet.
- Bad salespeople are less drawn to commission-based jobs.
- People with fire insurance are less likely to install smoke alarms.
- Brokers engage in risky market betting knowing they have corporate bailouts.

Discussion Activity 1

- How can you tell the difference between a customer who's just unlucky vs. one acting recklessly due to insurance?
- Think about incentives and behavior change. What would they have done without insurance?

Warranties and Moral Hazard

- Reckless behavior exists if warranties are provided as well (even more so)
- Consider a car company that allows purchase of an additional extended warranty of 1 year or 12,000 miles on a new car (a 10% failure rate). What can we expect out of the owners?
- First, consider the adverse selection problem. Those who will plan to drive carelessly everywhere are likely to purchase the car and use the warranty to cover any mechanical issues
- Next, consider the moral hazard problem. Once those who buy the car know they have the warranty, they will begin to drive recklessly.
- In both cases, the claim rate will exceed the failure rate.
- What can we do?
 - We price the warranty at greater than 10% of the car price. This allows us to hedge against claims.

Warranties and Moral Hazard

- Other ways to avoid/counter moral hazard with warranties:
 - Add clauses for improper use
 - Push full coverage warranties so consumers don't worry about costs
 - Promotes regular maintenance
 - Clauses may require the consumer to keep the product in the manufacturer's guidelines
 - Product registration and proof of purchase
 - Helps track usage
 - Helps reduce fraudulent claims
 - Professional installation
 - Failure to do so voids the warranty so more thought is given to the installation and upkeep.

Discussion Activity 2

- Why do companies sometimes require professional installation or registration to activate warranties?
- Consider how these requirements change behavior and help filter out risky users.

Shirking

- **Shirking** – a type of moral hazard that arises after an employee has been hired.
 - Shirking occurs when an employee doesn't work and is expected to.
- Under commission-based schemes, employees calculate their expected value of a sale.
 - If the expected value (probability times the commission) is greater than their cost to expend effort, then they will pursue the sale.
 - If the firm's contribution margin ($P - MC$) is equal to or less than the commission, then they cannot afford the sale.

Avoiding Shirking

- Incentive based compensation
 - Is a good idea if the cost to the company is less than the incentive to the individual.
 - The company would need to have enough of a contribution margin to cover the incentive to the shirker
 - If not, then it's in the company's best interest to not offer an incentive-based scheme
- Performance-Evaluation
 - A firm could invoke some form of monitoring to eliminate the shirking.
 - Off direct incentives: Reward (carrot) or penalize (stick) based on effort, not just outcomes
 - These types of performance metrics tie the employee directly to their performance to their compensation or their job (termination).

Avoiding Shirking

- Because monitoring can be relatively inexpensive it is beneficial to have more monitoring to eliminate shirking. Moreover, if an incentive plan doesn't aid in stopping shirking and doesn't solve the adverse selection and moral hazard problem use more monitoring.

Other ways to avoid shirking:

- Reputation
 - Establish a reputation of being a hard worker – can lead to higher wages.
 - Find workers who have established reputations
- Fixed fees
 - Accept part of the contract on a fixed fee, and the rest of the contract on an hourly based (commission + sales)

Activity – Warranties and Shirking

Scenario:

- You're a product manager for a fitness equipment company. Your company offers extended warranties and free replacements for parts that wear out. However, you've seen more customers returning machines with excessive damage and you suspect moral hazard.

Questions:

1. What evidence would suggest this is a moral hazard problem rather than bad luck?
2. Propose at least two contract clauses or policy changes that could reduce this problem.
3. How would you track whether your changes are effective?

Activity – Warranties and Shirking

1. Moral hazard signs: Unusual wear patterns, higher return rate among warranty buyers, decreased maintenance by customers
2. Contract solutions:
 - Require regular maintenance logs
 - Deny claims for misuse (e.g., excessive weight loads)
 - Tie extended warranties to product registration or training session
3. Effectiveness: Monitor claim frequency, cost per warranty, customer feedback, repair vs. replacement trends

Tools to Combat Moral Hazard

Category	Tool	Purpose / Example
Monitoring	Surveillance / Auditing	Monitor employee effort, enforce accountability
	Performance Reviews	Regular feedback discourages shirking
Incentive-Based	Commission / Bonuses	Align pay with output (e.g., sales jobs)
	Profit Sharing	Employees share in outcomes, reducing misaligned behavior
Contractual Design	Co-pays / Deductibles	Share cost to reduce overuse in insurance
	Covenants / Clauses in Loans	Limit risky actions post-loan
Ownership / Risk-Sharing	Equity Compensation	Employees have a stake in the company's long-term value
	Collateral Requirements	Borrowers put up assets to back loans

Discussion Activity 3

- What's a better approach to reduce shirking: paying commission or installing cameras? Why?
- Think about costs, trust, and employee motivation. What might work better in different jobs?

Turning Market Failure into Managed Risk

Moral hazard doesn't mean a deal shouldn't happen, it means we need to manage the risks.

Through smart contract design, monitoring, and incentives, we can:

- Anticipate risk-taking behavior
- Encourage responsible choices
- Protect both sides of the transaction

When structured well, moral hazard becomes a solvable problem, and the transaction can still create value.

Moral Hazard v. Adverse Selection Summary

Feature	Adverse Selection	Moral Hazard
When it occurs	Before the transaction	After the transaction
Problem type	Hidden information (unseen traits)	Hidden action (unseen behavior)
Main issue	Riskier parties are more likely to enter	Parties change behavior after gaining protection
Typical market effect	Pool gets riskier, good participants exit	Behavior gets riskier, system bears the cost
Health Insurance	Sick people are more likely to buy full coverage	People visit doctors more frequently once covered
Employment	Unqualified candidates apply for jobs with high salaries	Employees slack off once hired if not monitored
Lending	Risky borrowers more likely to seek credit	Borrowers invest in riskier projects after getting the loan
Solutions	Screening, signaling, tiered products	Monitoring, incentives, co-payments, contracts

Activity – Ways to Combat Moral Hazard

- Which of the following is not an example of a process designed to combat moral hazard problems?
 - Banks include restrictive covenants in loan agreements.
 - Universities have students complete evaluations of professor performance at the end of a class.
 - Insurance companies require applicants to provide medical history information as part of the application process.
 - Employers regularly monitor employee performance.

Moral Hazard and Lending

- **Adverse Selection**

- Problem: Higher-risk borrowers are more likely to seek loans
- Arises from: Hidden information about borrower's risk level

- **Moral Hazard**

- Problem: Borrowers may opt for riskier investments after securing a loan
- Example: Shift from a less risky investment to a much riskier one after receiving a loan

Example with Moral Hazard

- Initial loan for a less risky investment
 - Investment outcomes: \$100 or \$0 with equal probabilities
 - Bank's break-even on expected return.
- Shift to a riskier investment
 - Potential return: \$1,000 with 5% success chance
 - Result: Higher potential gain for borrower, increased risk for bank

Strategies to Mitigate Moral Hazard

- Monitoring borrower behavior
- Loan covenants to ensure funds are used as intended
- Requiring borrowers to invest their own money to share potential losses

Case Study: The 2008 Financial Crisis – Moral Hazard in Lending

Background

- Leading up to 2008, banks issued subprime mortgages to borrowers with low creditworthiness.
- These loans were bundled and sold as mortgage-backed securities.
- Institutions taking on the risk believed they'd be bailed out if things went wrong.
- Borrowers had little personal stake and often walked away from mortgages (strategic default).

Moral Hazard Factors

- Borrowers took riskier loans knowing foreclosure consequences were limited
- Mortgage originators made risky loans since they could sell them off
- Investors bought high-yield products thinking the government would step in if things collapsed
- Banks increased leverage and risky bets assuming they were “too big to fail”

Results

- Massive defaults, global financial contagion
- Firms like AIG, Bear Stearns, and Lehman Brothers collapsed or were bailed out
- Government bailouts prevented deeper collapse, but moral hazard worsened long-term

Key Takeaways

- Moral hazard occurs when people act differently after a deal because they don't bear the full consequences.
- Differs from adverse selection, which is about hidden information before a deal.
- Common settings:
 - Insurance: riskier behavior post-coverage
 - Employment: shirking after getting hired
 - Lending: borrowers taking on excessive risk
 - Warranties: reduced product care
- Solutions include:
 - Monitoring (cameras, evaluations, audits)
 - Incentives (performance pay, shared risk)
 - Contract clauses (usage restrictions, co-investment)
- Moral hazard doesn't mean the transaction shouldn't happen. It means risk needs to be anticipated and managed.

Knowledge Check

1. What is moral hazard, and how is it different from adverse selection?
2. Why might warranties encourage customers to take worse care of a product?
3. What is shirking, and why is it a moral hazard?
4. How can banks protect themselves from moral hazard after giving a loan?
5. Which of the following best reduces moral hazard?
 - A. Giving unconditional bonuses
 - B. Offering full warranties with no usage rules
 - C. Requiring co-investment from the borrower
 - D. Ignoring risky employee behavior

Knowledge Check Answers

1. Moral hazard = hidden actions after a deal (vs. hidden information before).
2. The customer knows repairs are covered, so they're less careful.
3. It's when employees underperform once hired, taking advantage of weak oversight.
4. Through monitoring, restrictive covenants, or requiring personal financial stake.
5. c) Requiring co-investment aligns incentives.

Consultant's Challenge: Managing Moral Hazard

Scenario:

- A high-growth fintech lender hires you to help reduce default rates on unsecured personal loans. They suspect borrowers are taking riskier actions post-funding and relying on lax repayment enforcement.

Your Task:

1. How would you identify whether the problem is moral hazard or adverse selection?
2. What mechanisms would you recommend to reduce risk after funding — without hurting good customers?
3. How could data, monitoring, or contract structure help align incentives?